THE GEOMETRY OF CATEGORICAL AND HIERARCHICAL CONCEPTS IN LARGE LANGUAGE MODELS

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ABSTRACT

The linear representation hypothesis is the informal idea that semantic concepts are encoded as linear directions in the representation spaces of large language models (LLMs). Previous work has shown how to make this notion precise for representing binary concepts that have natural contrasts (e.g., {`male`, `female`}) as *directions* in representation space. However, many natural concepts do not have natural contrasts (e.g., whether the output is about an animal). In this work, we show how to extend the formalization of the linear representation hypothesis to represent features (e.g., `is_animal`) as *vectors*. This allows us to immediately formalize the representation of categorical concepts as polytopes in the representation space. Further, we use the formalization to prove a relationship between the hierarchical structure of concepts and the geometry of their representations. We validate these theoretical results on the Gemma and LLaMA-3 large language models, estimating representations for 900+ hierarchically related concepts using data from WordNet.[1]

1 INTRODUCTION

Understanding how high-level semantic meaning is encoded in the representation spaces of large language models (LLMs) is a fundamental problem in interpretability. A particularly promising avenue is the *linear representation hypothesis* (e.g., Mikolov et al., 2013; Elhage et al., 2022; Nanda et al., 2023; Gurnee & Tegmark, 2024; Park et al., 2024). This is the informal hypothesis that semantic concepts are represented *linearly* in the representation spaces of LLMs. To assess the validity of this hypothesis, and systematically build tools on top of it, we must make precise what it means for a concept to be linearly represented, and understand how semantics are encoded in (the geometry of) the representation spaces.

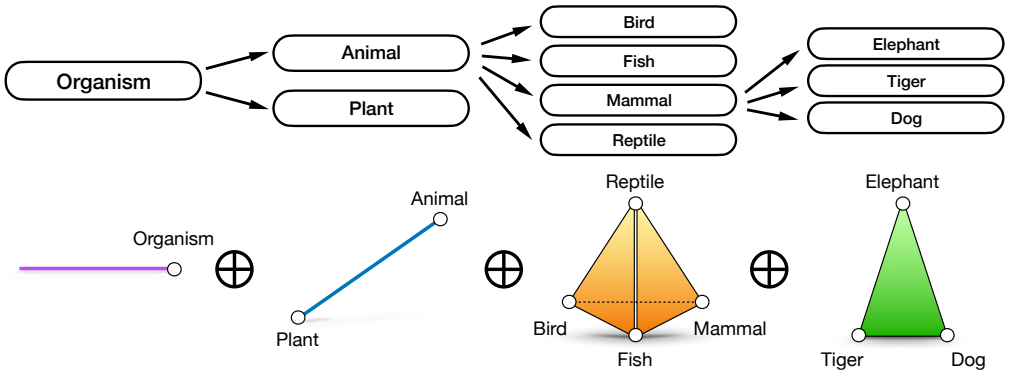
Focusing on the final softmax layer, Park et al. (2024) give a formalization for the case of binary concepts that can be defined by counterfactual pairs of words. For example, the concept `male` $\Rightarrow$ `female` is formalized using the counterfactual pairs {("man", "woman"), ("king", "queen"), ...}. They prove that such binary concepts have a well-defined linear representation as a direction in the representation space. They further connect semantic structure and representation geometry by showing that, under a suitable inner product, *causally separable* concepts that can be freely manipulated (e.g., `male` $\Rightarrow$ `female` and `french` $\Rightarrow$ `english`) are represented by orthogonal directions.

However, this formalization is limited: many natural concepts cannot be defined by counterfactual pairs of words. For example, simple binary features (e.g., `is_animal`) or categorical concepts (e.g., {`mammal`, `bird`, `reptile`, `fish`}) do not admit such formalizations. Additionally, it is not clear how semantic relationships beyond causal separability are encoded in the representation space. In particular, we are interested in this paper in understanding how hierarchical relationships between concepts are encoded in the representation space. That is, what is the relationship between the representations of `animal`, `mammal`, and `dog`?

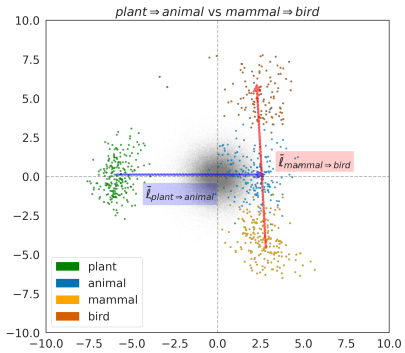
In this paper, we extend the linear representation hypothesis as follows:

1. We show how to move from representations of binary concepts as *directions* to representations as *vectors*. As a straightforward consequence, this allows us to represent categorical

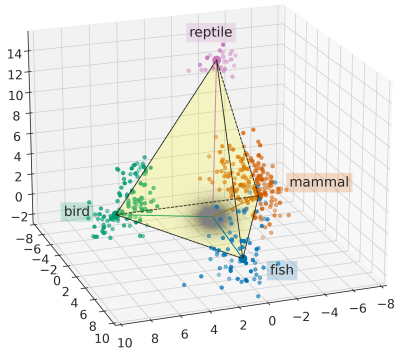
[1] Code is available at github.com/KihoPark/LLM_Categorical_Hierarchical_Representations.



(a) Pictorial depiction of the representation of hierarchically related concepts.



(b) Hierarchy is encoded as orthogonality in Gemma.



(c) Categorical concepts are represented as polytopes in Gemma.

Figure 1: In the representation spaces of LLMs, hierarchically related concepts (such as `plant` ⇒ `animal` and `mammal` ⇒ `bird`) live in orthogonal subspaces, while categorical concepts are represented as polytopes. The top panel illustrates the structure; the bottom panels show the measured representation structure in the Gemma LLM. See Section 5 and Appendix A for details.

concepts (e.g., {`mammal`, `bird`, `reptile`, `fish`}) as polytopes where each vertex is the vector representation of one of the elements of the concept (e.g., `is_bird`).

2. Using this result, we show that semantic hierarchy between concepts is encoded geometrically as orthogonality between representations, in a manner we make precise.
3. Finally, we empirically validate these theoretical results on the Gemma (Mesnard et al., 2024) and LLaMA-3 (Dubey et al., 2024) LLMs. To that end, we extract concepts from the WordNet hierarchy (Miller, 1995), estimate their representations, and show that the geometric structure of the representations aligns with the semantic hierarchy of WordNet.

The final structure is remarkably simple, and is summarized in Figure 1.

2 Preliminaries

We begin with some necessary background.

2.1 Large Language Models

For the purposes of this paper, we consider a large language model to consist of two parts. The first part is a function λ that maps an input text x to a vector $\lambda(x)$ in a representation space $\Lambda \simeq \mathbb{R}^d$. This is the function given by the stacked transformer blocks. We take $\lambda(x)$ to be the output of the final layer at the final token position. The second part is an unembedding layer that assigns a vector $\gamma(y)$ in an unembedding space $\Gamma \simeq \mathbb{R}^d$ to each token y in the vocabulary. Together, these define a

sampling distribution over tokens via the softmax distribution:

$$\mathbb{P}(y \mid x) = \frac{\exp(\lambda(x)^\top \gamma(y))}{\sum_{y' \in \text{Vocab}} \exp(\lambda(x)^\top \gamma(y'))}. \tag{2.1}$$

We note that the results that follow rely on the duality between the embedding and unembedding spaces, and the softmax link between them. Accordingly, we do not address the "internal" structure of the LLMs. However, we are optimistic that a clear understanding of the softmax geometry will shed light on this as well.

2.2 Concepts

We formalize a concept as a latent variable W that is caused by the context X and causes the output Y. That is, a concept is a thing that could—in principle—be manipulated to affect the output of the language model. In the particular case where a concept is a binary variable with a word-level counterfactual, we can identify the variable W with the counterfactual pair of outputs $(Y(0), Y(1))$. Concretely, we can identify `male` $\Rightarrow$ `female` with $(Y(0), Y(1)) \in_R \{$("man", "woman"), ("king", "queen"), ("he", "her"), $\dots\}$. We emphasize that the notion of a concept as a latent variable that affects the output is more general than the counterfactual binary case.

Given a pair of concept variables W and Z, we say that W is *causally separable* with Z if the potential outcome $Y(W = w, Z = z)$ is well-defined for all w, z. That is, two variables are causally separable if they can be freely manipulated—e.g., we can change the output language and the sex of the subject freely, so these concepts are causally separable.

2.3 Causal Inner Product and Linear Representations

We are trying to understand how concepts are represented. At this stage, there are two distinct representation spaces: Λ and Γ. The former is the space of context embeddings, and the latter is the space of token unembeddings. We would like to unify these spaces so that there is just a single notion of representation.

Park et al. (2024) show how to achieve this unification via a "Causal Inner Product". This is a particular choice of inner product that respects the semantics of language in the sense that the linear representations of (binary, counterfactual) causally separable concepts are orthogonal under the inner product. Their result can be understood as saying that there is some invertible matrix A and constant vector $\bar{\gamma}_0$ such that, if we transform the embedding and unembedding spaces as

$$g(y) \leftarrow A(\gamma(y) - \bar{\gamma}_0), \quad \ell(x) \leftarrow A^{-\top}\lambda(x) \tag{2.2}$$

then the Euclidean inner product in the transformed spaces is the causal inner product, and the Riesz isomorphism between the embedding and unembedding spaces is simply the usual vector transpose operation. We can estimate A as the whitening operation for the unembedding matrix. Following this transformation, we can think of the embedding and unembedding spaces as the same space, equipped with the Euclidean inner product.[2]

Notice that the softmax probabilities (eq. (2.1)) are unchanged for any A and $\bar{\gamma}_0$, so this transformation does not affect the model's behavior. The vector $\bar{\gamma}_0$ defines an origin for the unembedding space, and can be chosen arbitrarily. We give a particularly convenient choice below.

In this unified space, the linear representation of a binary concept $W \in_R \{0, 1\}$ is defined as:

Definition 1. A vector $\bar{\ell}_W$ is a linear representation of a binary concept W if for all contexts ℓ, and all concept variables Z that are causally separable with W, we have, for all $\alpha > 0$,

$$\mathbb{P}(W = 1 \mid \ell + \alpha\bar{\ell}_W) > \mathbb{P}(W = 1 \mid \ell), \text{ and} \tag{2.3}$$

$$\mathbb{P}(Z \mid \ell + \alpha\bar{\ell}_W) = \mathbb{P}(Z \mid \ell). \tag{2.4}$$

That is, the linear representation is a direction in the representation space that, when added to the context, increases the probability of the concept, but does not affect the probability of any off-target

[2] We are glossing over some technical details here; see Park et al. (2024) for details.